Knowledge Hub

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for the partial fulfilment of the requirements to award the degree of

Bachelor of Technology

In

Computer Science and Engineering

School of Engineering and Technology

Submitted by

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BML MUNJAL UNIVERSITY
FROM HERE TO THE WORLD

Under the Guidance of

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Certificate

Date: 11/28/2024

This is to certify that the project entitled **"Knowledge Hub System"** has been successfully carried out by the following students under my supervision:

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The work presented in this project is original, genuine, and fulfills the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering at BML Munjal University.

Supervisor
Dr. Nishtha Phutela

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Abstract

This project presents the design and implementation of a Knowledge **Hub System** using database management concepts. The system is developed to efficiently manage book records, user data, cart operations, and order processing in a structured and reliable manner. It addresses challenges such as data redundancy, inconsistent transactions, and inefficient querying in traditional systems.

The system is built using a relational database model with proper normalization up to Third Normal Form (3NF), ensuring data integrity and minimal redundancy. SQL queries including joins, grouping, and aggregation are used to retrieve meaningful insights, while transaction control ensures consistency during order placement. The results demonstrate improved performance, scalability, and accuracy in handling real-time operations, making the system suitable for practical applications.

Abbreviations

Short Form	Full Form
DBMS	Database Management System
ER	Entity Relationship
SQL	Structured Query Language
DDL	Data Definition Language
DML	Data Manipulation Language
DQL	Data Query Language
TCL	Transaction Control Language
PK	Primary Key
FK	Foreign Key
UI	User Interface
1NF	First Normal Form
2NF	Second Normal Form
3NF	Third Normal Form
ACID	Atomicity, Consistency, Isolation, Durability

List of Tables

For Address Table

Figure 1: Address Table Data

←T→	address_id	customer_id	full_name	phone	address_line	city	state	pincode	country	created_at	is_default
Edit Copy Delete	1	1	Akshat Bhardwaj	9155498563	Janki Sadan,House-30	Patna	Bihar	800020	India	2026-05-02 00:34:04	1
Edit Copy Delete	2	2	Akshat Bhardwaj	7070092920	Janki Sadan,House-31	Patna	Bihar	800020	India	2026-05-02 00:38:00	1
Edit Copy Delete	6	3	Akshat 2	9155498563	Janki Sadan,House-33	Patna	Bihar	800020	India	2026-05-02 06:16:45	1
Edit Copy Delete	7	3	Akshat 2	7070092921	BMI	Gurgaon	Haryana	122413	India	2026-05-02 06:17:19	0

For Books Table

Figure 2: Books Table Data

←T→	book_id	title	description	price	author	published_date	stock	category_id	image
Edit Copy Delete	12	The Alchemist	A philosophical novel about a shepherd named Santi...	399.00	Paulo Coelho	1988-01-01	9	1	https://m.media-amazon.com/images/S/compressed.pho...
Edit Copy Delete	13	Harry Potter	A young wizard begins his magical journey at Hogwa...	499.00	J.K. Rowling	1997-06-26	13	1	https://covers.openlibrary.org/b/id/7984916-L.jpg
Edit Copy Delete	14	Silence: Where Words Fall Short	This story is a quiet meditation on the irreversib...	699.00	Eric Matthes	2015-11-01	8	1	https://m.media-amazon.com/images/I/71K8j6r3PML._A...
Edit Copy Delete	15	Java: The Complete Reference	A complete reference covering Java programming con...	850.00	Herbert Schildt	2018-01-01	5	3	https://m.media-amazon.com/images/I/71Nm+UyZzpL._S...
Edit Copy Delete	16	A Brief History of Time	Explains the universe, black holes, time, and spac...	550.00	Stephen Hawking	1988-04-01	4	4	https://m.media-amazon.com/images/I/51X2RycvYSL._S...
Edit Copy Delete	17	The Selfish Gene	Explains evolution through gene-centered theory an...	480.00	Richard Dawkins	1976-01-01	6	4	https://m.media-amazon.com/images/I/51-blbzo1mL._S...
Edit Copy Delete	18	Sapiens	A brief history of humankind from evolution to mod...	600.00	Yuval Noah Harari	2011-01-01	9	4	https://m.media-amazon.com/images/I/713jloMO3UL._S...
Edit Copy Delete	19	Guns, Germs, and Steel	Explains how geography influenced civilizations an...	520.00	Jared Diamond	1997-03-01	9	4	https://m.media-amazon.com/images/I/61V8g4GgqdL._S...
Edit Copy Delete	20	World War 1: A History From Beginning to End	World War One was one of the bloodiest wars in mod...	750.00	Henry Freeman	2008-08-01	2	5	https://m.media-amazon.com/images/I/81yCmNYhj5L._A...
Edit Copy Delete	21	The Pragmatic Programmer	Tips and practices for becoming a better and more ...	700.00	Andrew Hunt	1999-10-30	6	3	https://covers.openlibrary.org/b/id/8091016-L.jpg
Edit Copy Delete	22	Deep Work	A guide to focused success in a distracted world.	650.00	Cal Newport	2016-01-01	9	5	https://m.media-amazon.com/images/I/71din4TLubL._A...
Edit Copy Delete	23	Atomic Habits	A guide to building good habits and breaking bad o...	550.00	James Clear	2018-01-01	9	5	https://m.media-amazon.com/images/I/71uEb+zNsoL._S...
Edit Copy Delete	24	Rich Dad Poor Dad	Lessons on money, investing, and financial indepen...	500.00	Robert Kiyosaki	1997-01-01	9	6	https://m.media-amazon.com/images/I/71HJj3XmheL._A...
Edit Copy Delete	25	The Lean Startup	Startup methodology for building products efficien...	600.00	Eric Ries	2011-01-01	7	5	https://m.media-amazon.com/images/I/81DjSfpjvNL._A...
Edit Copy Delete	26	Zero to One	Startup insights and innovation strategies.	700.00	Peter Thiel	2014-01-01	8	5	https://m.media-amazon.com/images/I/71uAI28kJuL._S...
Edit Copy Delete	27	The Psychology of Money	How people think about money and behavior.	520.00	Morgan Housel	2020-01-01	5	4	https://m.media-amazon.com/images/I/81gC3mdNi5L._S...
Edit Copy Delete	28	Introduction to Algorithms	Comprehensive guide to algorithms and data structu...	1200.00	Thomas Cormen	2009-01-01	3	1	https://m.media-amazon.com/images/I/9182pNXomXL._A...
Edit Copy Delete	29	Artificial Intelligence Basics	Introduction to AI and real-world applications.	800.00	Tom Taulli	2019-01-01	0	3	https://m.media-amazon.com/images/I/61m5AlecPTL._S...
Edit Copy Delete	30	Data Science Handbook	Guide to data science tools and techniques.	900.00	Field Cady	2017-01-01	1	3	https://m.media-amazon.com/images/I/713nxB9ZvlL._S...
Edit Copy Delete	31	Think and Grow Rich	Classic book on success mindset and wealth.	450.00	Napoleon Hill	1937-01-01	4	6	https://m.media-amazon.com/images/I/61IxJuRI39L._A...

For Category Table

Figure 3: Category Table Data

←T→	category_id	category_name
Edit Copy Delete	1	Fiction
Edit Copy Delete	5	History
Edit Copy Delete	3	Programming
Edit Copy Delete	4	Science
Edit Copy Delete	6	Technology

For Customer Table

Figure 4: Customer Table Data

←T→	customer_id	first_name	last_name	email	phone_number	password	role
Edit Copy Delete	1	Akshat	Bhardwaj	akshat.bhardwaj.24cse@bmu.edu.in	9155498563	$2y$10$TuyKMEStaletoV5iuyRj.ecAWCGUbgCxuQAXRPl15av...	user
Edit Copy Delete	2	Akshat	Bhardwaj	akshat12bhardwaj@gmail.com	7070092920	$2y$10$SYZdtExyBtztuIlzxA/KGuyozH8V2qa3BtcTkQHlQYJ...	admin
Edit Copy Delete	3	Akshat	2	akshat.2.24cse@bmu.edu.in	9155498563	$2y$10$49W8L24d4xDSrFv71tYUpud1FHYzRFSs.6ZzCsb/Vwy...	user

For Orders Table

Figure 5: Orders Table Data

←T→	order_id	order_date	total_amount	customer_id	order_status	is_archived	gst_amount	delivery_fee	address_id
Edit Copy Delete	1	2026-05-02 01:03:54	900.00	1	Pending	0	162.00	50.00	1
Edit Copy Delete	2	2026-05-02 04:24:36	450.00	1	Paid	0	81.00	50.00	1
Edit Copy Delete	3	2026-05-02 04:34:33	1200.00	1	Completed	0	216.00	0.00	1
Edit Copy Delete	4	2026-05-02 04:46:11	800.00	1	Paid	0	144.00	50.00	1
Edit Copy Delete	5	2026-05-02 04:47:09	520.00	1	Paid	0	93.60	50.00	1
Edit Copy Delete	6	2026-05-02 05:04:19	520.00	1	Paid	0	93.60	50.00	1
Edit Copy Delete	7	2026-05-02 05:21:41	520.00	1	Paid	0	93.60	50.00	1
Edit Copy Delete	14	2026-05-02 06:29:10	699.00	3	Placed	0	99.00	50.00	6
Edit Copy Delete	15	2026-05-02 06:32:01	520.82	3	Placed	0	71.82	50.00	6
Edit Copy Delete	16	2026-05-02 06:32:59	994.00	3	Placed	0	144.00	50.00	6
Edit Copy Delete	17	2026-05-02 06:59:01	663.60	1	Placed	0	93.60	50.00	1
Edit Copy Delete	18	2026-05-04 23:36:40	581.00	1	Placed	0	81.00	50.00	1

For Order_Items Table

Figure 6: Order_Items Table Data

←T→	order_id	book_id	quantity	subtotal
Edit Copy Delete	1	30	1	900.00
Edit Copy Delete	2	31	1	450.00
Edit Copy Delete	3	28	1	1200.00
Edit Copy Delete	4	29	1	800.00
Edit Copy Delete	5	27	1	520.00
Edit Copy Delete	6	27	1	520.00
Edit Copy Delete	7	27	1	520.00
Edit Copy Delete	14	23	1	550.00
Edit Copy Delete	15	12	1	399.00
Edit Copy Delete	16	29	1	800.00
Edit Copy Delete	17	27	1	520.00
Edit Copy Delete	18	31	1	450.00

For Payment Table

Figure 7: Payment Table Data

	payment_id	payment_method	payment_status	payment_date	order_id	amount
Edit Copy Delete	38	Card	Completed	2026-05-02 01:03:54	1	900.00
Edit Copy Delete	39	UPI	Completed	2026-05-02 04:24:36	2	450.00
Edit Copy Delete	40	COD	Completed	2026-05-02 04:34:33	3	1200.00
Edit Copy Delete	41	COD	Completed	2026-05-02 04:46:11	4	800.00
Edit Copy Delete	42	Card	Completed	2026-05-02 04:47:09	5	520.00
Edit Copy Delete	43	COD	Completed	2026-05-02 05:04:19	6	520.00
Edit Copy Delete	44	COD	Completed	2026-05-02 05:21:41	7	520.00
Edit Copy Delete	45	COD	Pending	2026-05-02 06:29:10	14	699.00
Edit Copy Delete	46	COD	Pending	2026-05-02 06:32:01	15	520.82
Edit Copy Delete	47	COD	Pending	2026-05-02 06:32:59	16	994.00
Edit Copy Delete	48	UPI	Pending	2026-05-02 06:59:01	17	663.60
Edit Copy Delete	49	COD	Pending	2026-05-04 23:36:40	18	581.00

For Wishlist Table

Figure 8: Wishlist Table Data

	wishlist_id	customer_id	book_id	created_at
Edit Copy Delete	11	1	20	2026-05-05 22:23:07
Edit Copy Delete	12	1	23	2026-05-05 22:23:10

List of Figures

For Address Table

Figure 1: Address Table Structure

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	address_id	int			No	None		AUTO_INCREMENT	Change Drop More
2	customer_id	int			No	None			Change Drop More
3	full_name	varchar(100)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
4	phone	varchar(15)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
5	address_line	varchar(255)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
6	city	varchar(100)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
7	state	varchar(100)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
8	pincode	varchar(10)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
9	country	varchar(50)	utf8mb4_0900_ai_ci		Yes	India			Change Drop More
10	created_at	timestamp			Yes	CURRENT_TIMESTAMP		DEFAULT_GENERATED	Change Drop More
11	is_default	tinyint(1)			Yes	0			Change Drop More

For Books Table

Figure 2: Books Table Structure

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	book_id	int			No	None		AUTO_INCREMENT	Change Drop More
2	title	varchar(100)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
3	description	text	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
4	price	decimal(10,2)			Yes	NULL			Change Drop More
5	author	varchar(100)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
6	published_date	date			Yes	NULL			Change Drop More
7	stock	int			Yes	NULL			Change Drop More
8	category_id	int			Yes	NULL			Change Drop More
9	image	varchar(255)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More

For Category Table

Figure 3: Category Table Structure

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	category_id	int			No	None		AUTO_INCREMENT	Change Drop More
2	category_name	varchar(100)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More

For Customer Table

Figure 4: Customer Table Structure

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	customer_id	int			No	None		AUTO_INCREMENT	Change Drop More
2	first_name	varchar(50)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
3	last_name	varchar(50)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
4	email	varchar(100)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
5	phone_number	varchar(15)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
6	password	varchar(100)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
7	role	varchar(10)	utf8mb4_0900_ai_ci		Yes	user			Change Drop More

For Orders Table

Figure 5: Orders Table Structure

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**order_id**	int			No	*None*		AUTO_INCREMENT	Change Drop More
2	**order_date**	datetime			No	CURRENT_TIMESTAMP		DEFAULT_GENERATED	Change Drop More
3	**total_amount**	decimal(10,2)			Yes	*NULL*			Change Drop More
4	**customer_id**	int			Yes	*NULL*			Change Drop More
5	**order_status**	varchar(40)	utf8mb4_0900_ai_ci		No	*None*			Change Drop More
6	**is_archived**	tinyint			Yes	0			Change Drop More
7	**gst_amount**	decimal(10,2)			No	0.00			Change Drop More
8	**delivery_fee**	decimal(10,2)			No	0.00			Change Drop More
9	**address_id**	int			Yes	*NULL*			Change Drop More

For Order_Items Table

Figure 6: Order_Items Table Structure

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**order_id**	int			No	*None*			Change Drop More
2	**book_id**	int			No	*None*			Change Drop More
3	**quantity**	int			Yes	*NULL*			Change Drop More
4	**subtotal**	decimal(10,2)			Yes	*NULL*			Change Drop More

For Payment Table

Figure 7: Payment Table Structure

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**payment_id**	int			No	*None*		AUTO_INCREMENT	Change Drop More
2	**payment_method**	varchar(50)	utf8mb4_0900_ai_ci		Yes	*NULL*			Change Drop More
3	**payment_status**	varchar(50)	utf8mb4_0900_ai_ci		Yes	*NULL*			Change Drop More
4	**payment_date**	datetime			No	CURRENT_TIMESTAMP		DEFAULT_GENERATED	Change Drop More
5	**order_id**	int			Yes	*NULL*			Change Drop More
6	**amount**	decimal(10,2)			No	*None*			Change Drop More

For Wishlist Table

Figure 8: Wishlist Table Structure

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**wishlist_id**	int			No	*None*		AUTO_INCREMENT	Change Drop More
2	**customer_id**	int			No	*None*			Change Drop More
3	**book_id**	int			No	*None*			Change Drop More
4	**created_at**	datetime			Yes	CURRENT_TIMESTAMP		DEFAULT_GENERATED	Change Drop More

1. Introduction

With the rapid growth of digital technology, online platforms have become a primary medium for purchasing products, including books. Online bookstores provide users with convenience, a wide selection of books, and easy access to information. However, managing large amounts of data such as books, customers, orders, and payments requires an efficient and well-structured system.

A Database Management System (DBMS) plays a key role in organizing, storing, and retrieving data while ensuring consistency, reducing redundancy, and supporting efficient query processing.

The **Knowledge Hub System** is designed to manage book listings, user data, cart operations, and order processing through a relational database. The system ensures accurate data handling, efficient transactions, and improved user experience.

Problem Statement

Managing an online bookstore involves handling large volumes of data such as books, customers, orders, payments, and inventory. Traditional or poorly structured systems often face issues like data redundancy, inconsistent records, inefficient query processing, and difficulty in maintaining relationships between different entities.

The problem is to design and implement a structured database system that can efficiently store, manage, and retrieve this data while ensuring data integrity, scalability, and reliable transaction handling. The system should support real-time operations such as browsing books, managing cart items, placing orders, and processing payments.

The **Knowledge Hub System** aims to address these challenges by providing a well-designed relational database with proper normalization, efficient query execution, and transaction management to ensure accuracy and performance.

2. Methodology

2.1.Design (ER Diagram, Schema, Normalization)

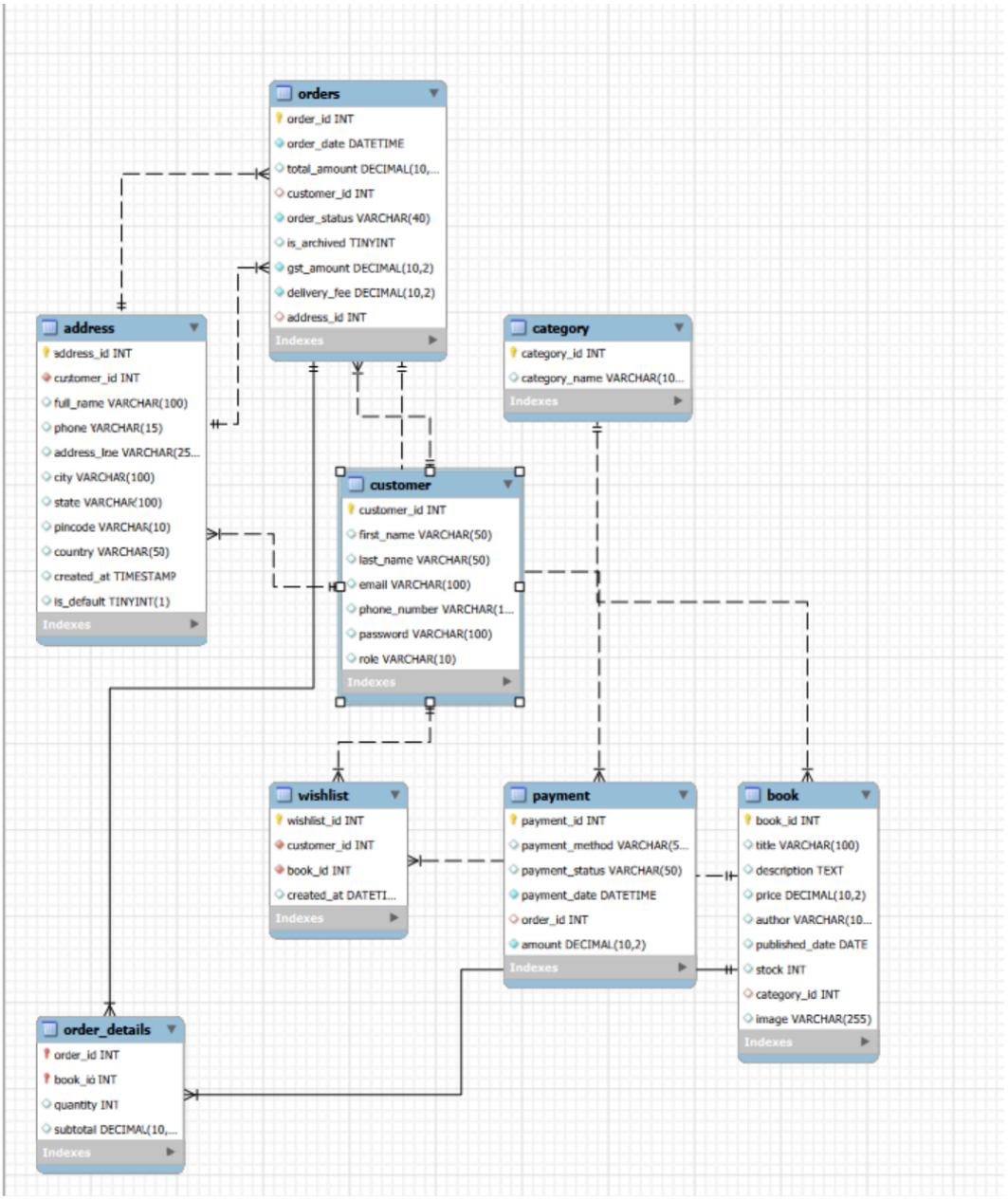

Figure 9: ER Diagram of Knowledge Hub System

The above figure represents the Entity-Relationship (ER) diagram of the Knowledge Hub System, illustrating the key entities, their attributes, and relationships within the database. The Customer entity stores user details such as name, email, and login credentials, and is connected to the Orders entity through a one-to-many relationship, as a single customer can place multiple orders.

The Orders entity maintains order-specific information including order date, total amount, and order status. Each order is linked to one customer and can contain multiple books. The Book entity stores details such as title, price, author, and stock, and is associated with the Category entity through a many-to-one relationship for proper classification of books.

The many-to-many relationship between Orders and Books is resolved using the Order_Details (Order_Items) table, which stores attributes such as quantity and subtotal for each book in an order. The Payment entity is linked to Orders and manages transaction-related information including payment method, payment status, and amount.

Additionally, the Wishlist entity allows customers to save books for future reference, forming a relationship between customers and books. Overall, the ER diagram provides a structured and normalized representation of the database, ensuring data integrity, minimizing redundancy, and enabling efficient query processing.

2.2.Implementation

The database is implemented using SQL in a relational database management system (MySQL). Tables are created using DDL commands with appropriate constraints such as primary keys, foreign keys, and default values.

Data is inserted and managed using DML queries, and data retrieval is performed using DQL queries including joins, grouping, and filtering operations.

The system is integrated with a web interface that allows users to browse books, add items to the cart, and place orders. Transactions are used during order placement to ensure consistency, using COMMIT for successful operations and ROLLBACK in case of failures.

Indexing is applied on frequently accessed columns such as book_id and customer_id to improve query performance.

2.3.Algorithm:

Input:

Customer selection of books and quantity

Output:

Order successfully placed with updated database records

Step 1: *User selects books and adds them to the cart*

Step 2: *User proceeds to checkout and enters address details*

Step 3: *System calculates total amount including taxes and delivery charges*

Step 4: *Order record is created in the Orders table*
Step 5: *Selected books are inserted into Order_Items table*
Step 6: *Payment details are recorded in the Payment table*
Step 7: *Transaction is committed if all operations are successful*
Step 8: *If any error occurs, transaction is rolled back*

3. Discussion/Results/Observations

3.1 Database Schema and Query Results

The following results demonstrate the implementation of theKnowledge Hub System. The database schema has been successfully created with multiple interconnected tables such as Customer, Books, Orders, Order_Items, Payment, Address, Category, and Wishlist. These tables are linked using primary and foreign keys to maintain relationships and ensure referential integrity.

Various SQL queries were executed to retrieve and manipulate data, including SELECT, JOIN, and GROUP BY operations. The system supports real-time functionalities such as adding items to the cart, placing orders, and processing payments. Transaction control using COMMIT and ROLLBACK ensures consistency and reliability of data. Overall, the results show efficient data handling and optimized query execution.

order_id	first_name	title	quantity	subtotal
1	Akshat	Data Science Handbook	1	900.00
2	Akshat	Think and Grow Rich	1	450.00
3	Akshat	Introduction to Algorithms	1	1200.00
4	Akshat	Artificial Intelligence Basics	1	800.00
5	Akshat	The Psychology of Money	1	520.00
6	Akshat	The Psychology of Money	1	520.00
7	Akshat	The Psychology of Money	1	520.00
17	Akshat	The Psychology of Money	1	520.00
18	Akshat	Think and Grow Rich	1	450.00
14	Akshat	Atomic Habits	1	550.00
15	Akshat	The Alchemist	1	399.00
16	Akshat	Artificial Intelligence Basics	1	800.00

Figure 10: Query Output Showing Orders, Customers, and Book Details

The above query output displays the relationship between orders, customers, and books using JOIN operations. It shows details such as order ID, customer name, book title, quantity, and subtotal. This demonstrates how data from multiple tables is combined to generate meaningful results.

Table 1: Sample Query Output

order_id	first_name	title	quantity	subtotal
1	Akshat	Data Science Handbook	1	900
2	Akshat	Think and Grow Rich	1	450

4. Concluding Remarks

The Knowledge Hub System successfully demonstrates the application of database management concepts in a real-world scenario. The system efficiently manages data

related to books, customers, orders, and payments using a structured relational database.

The use of normalization ensures reduced redundancy and improved data consistency, while SQL queries such as joins and aggregations enable efficient data retrieval. Transaction management ensures reliable order processing.

Overall, the system provides a scalable and efficient solution for managing online bookstore operations and highlights the practical importance of DBMS in modern applications.

5. Any societal relevance of your work

The system contributes to society by making book purchasing more accessible and convenient through an online platform. It allows users to browse and order books from anywhere, promoting digital literacy and easy access to educational resources.

Additionally, it helps businesses manage their operations efficiently, reducing manual effort and improving accuracy in handling orders and transactions.

6. Future scope (how would you like to expand this, if any plan)

The system can be further enhanced by adding features such as recommendation systems based on user preferences, advanced search filters, and real-time order tracking.

Integration with secure payment gateways, mobile applications, and cloud-based databases can improve scalability and user experience. Additionally, implementing analytics and reporting features can help in better decision-making for business growth.

7. Team details and individual contribution

Akshat Bhardwaj – Technical Lead
Handled overall system integration, backend logic, and key features like cart, checkout, orders, and transactions. Ensured data consistency and debugging.

Abhishek Rana – UI & Customer Experience
Developed frontend pages including browsing, cart, and checkout. Focused on usability and UI–backend integration.

Bhanu – Database Management
Designed database schema, normalization, and implemented core tables and SQL queries.

Ali – Admin & Inventory
Developed admin panel features, managed book inventory, and handled order monitoring.

Amit – Authentication & Security
Implemented login, registration, session handling, and user authentication.

References

1. Silberschatz, A., Korth, H. F., & Sudarshan, S. *Database System Concepts*, McGraw-Hill, 6th Edition.
2. Elmasri, R., & Navathe, S. B. *Fundamentals of Database Systems*, Pearson, 7th Edition.
3. MySQL Documentation – https://dev.mysql.com/doc/
4. PHP Official Documentation – https://www.php.net/docs.php
5. W3Schools SQL Tutorial – https://www.w3schools.com/sql/
6. GeeksforGeeks – SQL and DBMS Concepts – https://www.geeksforgeeks.org/dbms/

Declaration of AI Usage

Artificial Intelligence (AI) tools were used in this project only for learning assistance, debugging, documentation support, and improving code structure. AI helped in understanding DBMS concepts such as normalization, joins, transactions, stored procedures, and query optimization.

The final database design, SQL implementation, frontend integration, testing, and overall project development were performed manually by the developer. All AI-generated suggestions were reviewed, modified, and verified before being included in the project.

AI was used as a supportive educational tool and not as a replacement for original development work.